Deepan Chandrasekaran

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE student (CGPA 9.15) with hands-on experience building and deploying computer vision, deep learning, and GenAI systems. Proficient in PyTorch, CNNs, OpenCV, and Python. Built production-grade image classification models (99.8% accuracy), real-time inference pipelines (45 FPS), and RAG-based GenAI platforms. Strong foundation in probability, linear algebra, optimization, and model interpretability. National hackathon winner.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C/C++

Computer Vision & Deep Learning: PyTorch, CNNs, OpenCV, Image Classification, Feature Extraction, Model Interpretability, Real-Time Inference, Edge Deployment, Transfer Learning, Object Recognition

ML & GenAI: Hugging Face, Sentence Transformers, RAG Pipelines, FAISS, ChromaDB, LLM Integration (OpenAI, Gemini, Ollama/Mistral), Prompt Engineering, Fine-tuning, Vector Search

Frameworks & Tools: FastAPI, React, Spring Boot, Docker, Git, Linux, AWS (basics), NVIDIA A100/H100/T4

Mathematics: Probability & Statistics, Linear Algebra, Calculus, Geometry, Optimization

PROJECTS

ISL Sign Language Recognition – Computer Vision & Deep Learning | *Aug – Nov 2025* | *PyTorch, CNNs, OpenCV, Python, NVIDIA A100/H100/T4*

- Designed and trained a CNN-based image classification model on 42,000 labeled images achieving **99.8% validation accuracy** and 90% test accuracy; architecture scalable to 1M+ samples on HPC GPUs.
- Achieved **real-time inference at 45 FPS** on edge devices; applied model optimization and edge deployment techniques directly analogous to item authentication and fingerprinting use cases.
- Applied feature extraction, transfer learning, and data augmentation; analyzed model interpretability using activation maps to understand visual characteristics driving predictions.

LaRa – Real-Time AI Platform with Deep Learning and GenAI | *Jan 2026 – Present* | *PyTorch, FastAPI, Faster-Whisper, RAG, Docker, WebRTC VAD*

- **Won 3rd Place at RAMPAGE V26, National Level Hackathon (KL-H).** Production-grade system integrating LLM narrative generation, deterministic FSM decision engine, and RAG-based long-term memory.
- Built a real-time speech pipeline using Faster-Whisper and WebRTC VAD with deep learning inference; containerized with Docker for production deployment; implemented secure offline processing with no cloud dependency.
- Engineered modular, well-tested APIs (/health, /start, /stop) with async FastAPI endpoints; wrote clean, unit-tested code following software engineering best practices.

ResumeAnalyse – Semantic Search and RAG Pipeline | *Dec 2025 – Feb 2026* | *Sentence Transformers, ChromaDB, Ollama, Mistral, React, Python*

- Built an end-to-end RAG pipeline: PDF chunking, Sentence Transformer embeddings, ChromaDB vector DB, semantic retrieval; achieved **>90% job-resume relevance** matching accuracy.
- Integrated Mistral LLM via Ollama for context-grounded Q&A with prompt engineering; reduced recruiter analysis time by 50%; deployed secure RESTful backend consumed by React frontend.

Search Wizard – Multimodal Semantic Search Engine | *2025* | *Python, FAISS, Gemini Embeddings, BM25, FastAPI, Electron*

- Multimodal semantic search over 50K+ files using FAISS (768-dim) and BM25 hybrid retrieval; FastAPI backend with RAG-style document Q&A powered by Gemini Flash LLM.
- Open-source cross-platform Electron app with real-time indexing and vector search; contributed full documentation and clean modular code for community use.

WORK EXPERIENCE

Software Developer (Freelance) – Local Apparel Manufacturer | *2023*

- Engineered a full-stack shirt customization platform (React, Three.js) with 3D product preview and order management backend; applied full SDLC from requirements through deployment; increased client sales by 8-10%.
- Resolved third-party API integration defects under delivery constraints; gathered and acted on end-user feedback to continuously improve software reliability and usability.

EDUCATION AND CERTIFICATIONS

B.Tech in Computer Science and Engineering – KL University, Vijayawada | Expected April 2027 | CGPA: 9.15/10

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Computer Networks, Software Engineering

Oracle Cloud Infrastructure 2025 – Generative AI Professional | Oracle University | Oct 2025 – Oct 2027

Oracle AI Vector Search Professional | Oracle University | Oct 2025 – Oct 2027

CODING PROFILES

LeetCode: leetcode.com/u/kl2300032731 | **CodeChef:** codechef.com/users/klu2300032731